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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction to Graph Algorithms (course)

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Course
outline

About
NPTEL ()

How does an
NPTEL
online
course
work? ()

Week 1
Introduction
and
Principles of
Algorithms ()

Week 2
Shortest
Path
Algorithms
and
problems ()

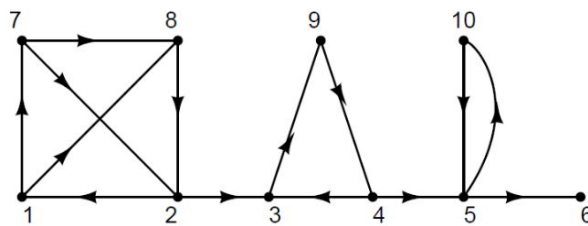
Week 3
Bellman
Equations ()

Week 8 Assignment 8

Your last recorded submission was on 2025-09-17, 09:56 Due date: 2025-09-17, 23:59 IST.
IST

1) Let G be the ___ for the given graph.

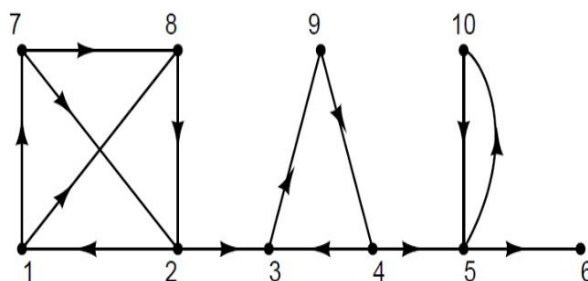
2 points



The vertices in the largest strongly connected components are { }

- ☐ {1,2,7,8}
☐ {1,2,3,7,8}
☐ {1,2,5,8}
☒ {1,2,5,7,8}

2) Let G be the ___ for the given graph.



The number of strongly connected components are ___

4

2 points

Week 4
Bellman
Ford and
Dijkstra
Algorithm ()

Week 5 All
Pair Shortest
Path ()

Week 6
Prims and
Kruskal's
Algorithm ()

Week 7 DFS
and Cut
Vertex ()

Week 8
Strong
Connected
Components
& BFS ()

☐ Strong
 Connected
 Components
 (unit?
 unit=25&lesso
 n=87)

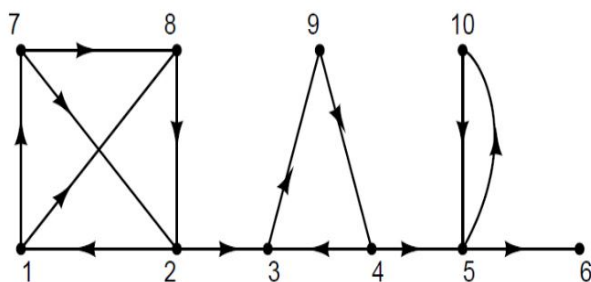
☐ Strong
 Connected
 Components
 Part 2 (unit?
 unit=25&lesso
 n=88)

☐ Strong
 Connected
 Components
 Part 3 (unit?
 unit=25&lesso
 n=89)

☐ Strong
 Connected
 Components
 Part 4 (unit?
 unit=25&lesso
 n=90)

☐ BFS (unit?
 unit=25&lesso

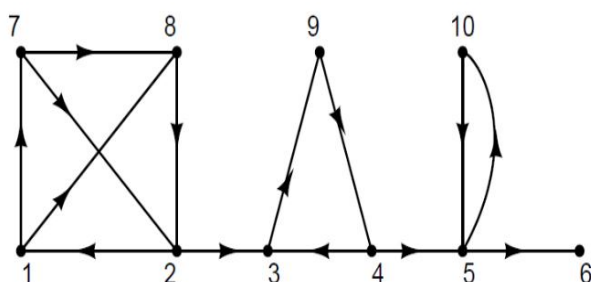
3) Let G be the ___ for the given graph.



The number of edges in the condensed DAG is ___

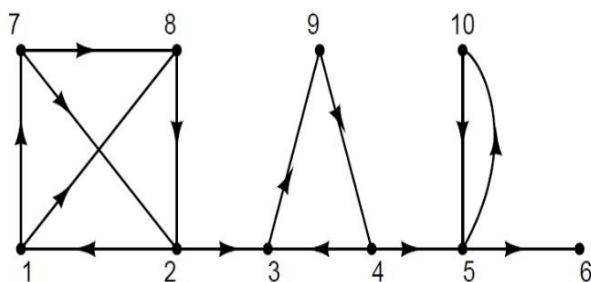
2 points

4) Suppose we do a BFS in the given graph starting from vertex 1, then the number of levels are (assuming root is at level 0) ___.



2 points

5) Suppose we do a BFS in the given graph G starting from vertex 2, then the number of levels in the BFS tree is ___.



2 points

You may submit any number of times before the due date. The final submission will be considered for grading.

Submit Answers

n=91)

☐ Week 8
Lecture
Material (unit?
unit=25&lesso
n=69)

☒ **Quiz: Week 8
Assignment 8
(assessment?
name=104)**

☐ Feedback
Form (unit?
unit=25&lesso
n=33)

**Video
download ()**

**Live session
()**

